

## DIB FORMAT FOR THE PROJECT IICT-CS

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| <b>1</b>  | <b>Project Outline</b>                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.1       | Title of the Project                                                                                                                                                                         | Chemical Sciences and Technologies for Health, Agriculture and Environment (IICT-CS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.2       | Sponsoring Agency                                                                                                                                                                            | Ministry: Ministry of Science & Technology<br>Department: Department of Scientific & Industrial Research (DSIR)<br>Autonomous Body: Council of Scientific & Industrial Research (CSIR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.3       | Proposed Cost of the Project                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.4       | Proposed Timelines for the Project                                                                                                                                                           | 5 Years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1.5       | Whether Project will be Implemented as a part of Scheme or on Stand-Alone Basis?                                                                                                             | As a part of the National Laboratories Scheme of DSIR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1.6       | Whether financial resources required for the project have been tied up? If yes, details                                                                                                      | No. Is part of the EFC on National Laboratories Scheme being submitted to DoE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.7       | Whether Feasibility Report and/or Detailed Project Report has been Prepared?                                                                                                                 | Detailed Project Report Prepared                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.8       | Whether the proposal is an Original Cost Estimate or a Revised Cost Estimate                                                                                                                 | Original Cost Estimate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.9       | In case of Revised Cost Estimates, whether the meeting of Revised Cost Estimate has been held and its recommendations suitably addressed                                                     | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.10      | Whether any land acquisition or pre-investment activity was undertaken or is contemplated for this Project? Whether the cost of such intervention has been included in the Project Proposal? | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>2.</b> | <b>Outcomes and Deliverables</b>                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1       | Stated Aims and Objectives of the Project                                                                                                                                                    | CSIR-IICT, with over 80 years of excellence in chemical sciences, is well positioned to contribute meaningfully to India's aspirations in healthcare, agriculture, energy, and environmental sustainability. As a premier chemical technology innovation institute, IICT's established strengths in organic synthesis, catalysis, fluorine chemistry, chemical biology, materials science, biotechnology, chemical and environmental engineering, align strongly with national priorities such as Atmanirbhar Bharat, and the Sustainable Development Goals. India's rapidly growing demand for advanced pharmaceuticals, fluoro and agrochemicals, specialty chemicals, and clean energy solutions involving indigenous innovation, value creation. CSIR-IICT's strong record of translational achievements including indigenous Hydrazine Hydrate production, process for adjuvant used in covaxin, and industry-adopted process technologies demonstrates its ability to convert cutting-edge research into scalable assets. Supported by core strengths in chemistry and chemical technology along with excellent infrastructure and multidisciplinary scientific expertise, this project aims at sustainable drug discovery and API processes, market- |

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|                                                                                                                                                |                                                                           | <p>driven fluoro, agro, specialty chemical and oil technologies, waste-to-wealth based circular economy solutions, clean water technologies and functional materials for energy harvesting and storage. The institution is strategically advancing research in Carbon Capture and Utilization (CCU) with the objective of developing low-energy, compact, and scalable CO<sub>2</sub> separation systems both from point sources (biogas, cement, steel, power plant flue gases) as well as directly from air suitable for industrial deployment. Collectively these integrated research thrusts reinforce CSIR-IICT's core mandate under Vision-2032 by enabling sustainable chemical technologies that enhance national self-reliance, strengthen industrial competitiveness, promote environmental stewardship, and deliver societal and economic benefits across multiple sectors</p> <p><b>Objective of the Project:</b></p> <ul style="list-style-type: none"><li>• Discovery and development of novel therapeutic molecules and processes toward pharmaceuticals using sustainable approaches.</li><li>• Development of process technologies for market driven fluorochemicals, specialty chemicals, agrochemicals and pheromones.</li><li>• Develop integrated waste-to-wealth technologies for converting solid and liquid waste into clean energy (biogas/CBG), platform chemicals, and high-value products, advancing India's circular economy transition.</li><li>• Development of CO<sub>2</sub> capture and utilization technologies converting emissions into fuels and chemicals.</li><li>• Development of advanced functional materials and composites for efficient energy harvesting &amp; storage applications and oil technologies.</li></ul> |         |         |         |                                |
| 2.2                                                                                                                                            | Indicate year-wise outputs/deliverables for the project in a tabular form |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |         |         |                                |
| Healthcare                                                                                                                                     |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |         |         |                                |
| Deliverables                                                                                                                                   | 2026-27                                                                   | 2027-28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2028-29 | 2029-30 | 2030-31 | Overall five-year deliverables |
| Isolation of bioactive secondary metabolites<br>Marker compound identification and standardization of herbal drugs and polyherbal formulations | 5-8                                                                       | 10-12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 10-12   | 10-12   | 10-12   | 45-60                          |

|                                                                                                      |       |       |       |       |       |         |
|------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|---------|
| Structure elucidation of novel natural products                                                      | 5-8   | 10-12 | 10-12 | 10-12 | 10-12 | 10-15   |
| Chemical fingerprinting and quality control of extracts/formulations using HPLC, HPTLC, LC-MS        | 1     | 1-3   | 1-3   | 1-3   | 1-3   | 5-15    |
| Phytopharmaceutical leads                                                                            | 0     | 1     | 1     | 1     | 1     | 2-4     |
| Publications                                                                                         | 1     | 2     | 2     | 2     | 2     | 8-10    |
| Patents                                                                                              | 0     | 0     | 1     | 1     | 1     | 3-5     |
| Structure-based drug design and molecular modelling for lead identification and optimization         | 10-15 | 15-20 | 15-20 | 15-20 | 15-20 | 70-80   |
| Synthesis of focused molecular libraries and scaffold diversification for SAR generation             | 10-15 | 20-25 | 20-25 | 20-25 | 20-25 | 90-120  |
| Development of peptidomimetics and peptide-based therapeutics with improved stability and potency.   | 1     | 3-4   | 3-4   | 3-4   | 3-4   | 12-16   |
| Design and development of photo responsive drug delivery systems for controlled release applications | 1     | 1-2   | 1-2   | 1-2   | 1-2   | 4-8     |
| Synthesis of fluorescent molecular probes for imaging, diagnostics, and target engagement studies    | 1     | 1-2   | 1-2   | 1-2   | 1-2   | 4-8     |
| Publications                                                                                         | 10-20 | 30-50 | 30-50 | 30-50 | 30-50 | 130-250 |
| Patents                                                                                              | 2-6   | 4-6   | 4-6   | 4-6   | 4-6   | 18-30   |
| HR Training                                                                                          | 20    | 20    | 20    | 20    | 20    | 100     |
| Synthesis of potent biologically active natural products and their analogues towards the             | 10-12 | 10-15 | 10-15 | 10-15 | 10-15 | 50-75   |

|                                                                                                         |                |                |                |                |                |                                        |
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| development of potent molecules including peptides, nucleosides and nucleotides.                        |                |                |                |                |                |                                        |
| New libraries of therapeutically relevant new chemical entities.                                        | 10             | 20             | 20             | 20             | 20             | 50-100                                 |
| Green activation strategies using electro- & photochemistry and flow-chemistry                          | 1-2            | 2-4            | 2-4            | 2-4            | 2-4            | 10-16                                  |
| Scalable continuous-flow processes for Key starting materials towards Drug Substances.                  | 1-2            | 2-4            | 2-4            | 2-4            | 2-4            | 9-18                                   |
| Screening for anticancer activity, and Structure-activity relationship.                                 | 500            | 1000           | 1000           | 1000           | 1000           | 4500                                   |
| Screening of molecules for neurotrophic activity and Structure-activity relationship.                   | 500            | 1000           | 1000           | 1000           | 1000           | 4500                                   |
| Screening of molecules for anti-bacterial and anti-fungal activity and Structure-activity relationship. | 1000           | 1000           | 1000           | 1000           | 1000           | 5000                                   |
| Potential hits                                                                                          | 10-15          | 10-15          | 10-15          | 10-15          | 10-15          | 50-75                                  |
| publications                                                                                            | 3-5            | 5-8            | 5-8            | 5-8            | 5-8            | 25-40                                  |
| Patents                                                                                                 | 0              | 0              | 1              | 1              | 1              | 3-5                                    |
| HR Training                                                                                             | 4-5            | 4-5            | 4-5            | 4-5            | 4-5            | 20-25                                  |
| In vivo validation of HIT compounds                                                                     | 0              | 1-2            | 2-3            | 2-3            | 2-3            | 7-12                                   |
| Bioavailability of Hit Compounds                                                                        | 0              | 1-2            | 2-3            | 2-3            | 2-3            | 7-12                                   |
| HR Training                                                                                             | 4-5            | 4-5            | 4-5            | 4-5            | 4-5            | 20-25                                  |
| <b>Agrochemicals</b>                                                                                    |                |                |                |                |                |                                        |
| <b>Deliverables</b>                                                                                     | <b>2026-27</b> | <b>2027-28</b> | <b>2028-29</b> | <b>2029-30</b> | <b>2030-31</b> | <b>Overall, five-year deliverables</b> |
| Process know-how for fluoro and agrochemicals                                                           | 1              | 1              | 1              | 1-2            | 1-2            | 5-7                                    |
| Pilot scale process for tech transfer                                                                   | 0              | 1              | 1              | 1              | 0              | 3                                      |
| Potential techno-commercial packages of                                                                 | 0              | 0              | 0              | 1              | 1              | 2                                      |

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| agrochemicals for out licensing                                                                                                                              |                  |                  |                |                |                   |                                       |
| Process technologies for next generation HFOs                                                                                                                | 0                | 1                | 1              | 1              | 0                 | 3                                     |
| Generation of library of designed new molecules with as agrochemicals with potential for further development; Hit generation to lead optimization            | 10 hit molecules | 20 hit molecules | 5 leads        | 20 leads       | 5 optimized leads | 30 Hit molecules<br>25 leads          |
| <b>Waste to Wealth</b>                                                                                                                                       |                  |                  |                |                |                   |                                       |
| <b>Deliverables</b>                                                                                                                                          | <b>2026-27</b>   | <b>2027-28</b>   | <b>2028-29</b> | <b>2029-30</b> | <b>2030-31</b>    | <b>Overall five-year deliverables</b> |
| <b>Technology:</b><br>Anaerobic Gas Lift Reactor (AGR) and Dry Anaerobic Digester with unique Hydrodynamics (DADUH) for the Treatment of Organic Solid Waste | 1                | 1                | 1              | 1-2            | 1-2               | 4-6                                   |
| Bi-phasic Anaerobic Digestion System for High Methane Biogas Production                                                                                      | 1                | 1                | 1              | 1-2            | 1-2               | 4-6                                   |
| Accelerated Anaerobic Composting (AAC) of Organic Waste                                                                                                      | 1                | 1                | 1              | 1-2            | 1-2               | 4-6                                   |
| Green Hydrogen Production in Biorefinery Platform                                                                                                            | 1                | 1                | 1              | 1-2            | 1-2               | 4-6                                   |
| Biosuccinic acid Production through Hybrid Fermentation Route- Bioprocess Development                                                                        | 1                | 1                | 1              | 1-2            | 1-2               | 4-6                                   |
| Co-fermenting Lactic Acid and Glucose towards Caproic Acid Production                                                                                        | 1                | 1                | 1              | 1-2            | 1-2               | 4-6                                   |
| <b>Carbon Capture, Utilization &amp; Storage (CCUS):</b>                                                                                                     |                  |                  |                |                |                   |                                       |
| <b>Deliverables (Carbon Capture)</b>                                                                                                                         | <b>2026-27</b>   | <b>2027-28</b>   | <b>2028-29</b> | <b>2029-30</b> | <b>2030-31</b>    | <b>Overall five-year</b>              |

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|                                                                                                                                                       |                |                |                |                |                | <b>deliverables</b>                   |
| Development of candidate solvents kinetic data generation                                                                                             | 3              | 3              | 0              | 0              | 0              | 6                                     |
| Optimization of material systems showing improved CO <sub>2</sub> capture intensified absorber/contacter with experimental verification.              | 2              | 2              | 0              | 0              | 0              | 4                                     |
| Continuous operation of capture system at lab scale                                                                                                   | 0              | 0              | 3              | 0              | 0              | 3                                     |
| Fabrication and commissioning of pilot-scale integrated system; long-duration operation (>500 h) and detailed TEA and LCA for performance validation. | 0              | 0              | 1              | 2              | 0              | 3                                     |
| Demonstration of stable pilot-scale operation                                                                                                         | 0              | 0              | 0              | 0              | 3              | 3                                     |
| <b>Deliverables (Carbon Utilization)</b>                                                                                                              | <b>2026-27</b> | <b>2027-28</b> | <b>2028-29</b> | <b>2029-30</b> | <b>2030-31</b> | <b>Overall five-year deliverables</b> |
| Development of catalysts/material systems with full characterization and performance testing                                                          | 3              | 2              | 2              | 0              | 0              | 7                                     |
| Process optimization studies for improved conversions and yields, modelling and Design of catalytic reactor with experimental verification.           | 2              | 2              | 2              | 0              | 0              | 6                                     |
| Continuous operation (>100 h) CC system at lab scale; completion of pilot-scale design with preliminary TEA and LCA reports.                          | 0              | 0              | 2              | 1              | 0              | 3                                     |
| Fabrication and commissioning of                                                                                                                      | 0              | 0              | 0              | 2              | 0              | 2                                     |

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| pilot-scale integrated system; long-duration operation (>500 h) and detailed TEA and LCA for performance validation.     |                |                |                |                |                |                                       |
| Demonstration of stable and optimal pilot-scale operation                                                                | 0              | 0              | 0              | 0              | 2              | 2                                     |
| <b>Specialty Chemicals:</b>                                                                                              |                |                |                |                |                |                                       |
| <b>Deliverables</b>                                                                                                      | <b>2026-27</b> | <b>2027-28</b> | <b>2028-29</b> | <b>2029-30</b> | <b>2030-31</b> | <b>Overall five-year deliverables</b> |
| Process technologies for 5 specialty chemicals (lab scale development) Catalyst design, screening, and stability studies | 2              | 2              | 1              | 1              | 1              | 5-7                                   |
| Development of intensified continuous reactor platforms                                                                  | 0              | 2              | 2              | 1              | 1              | 4-6                                   |
| Modular intensified reactor prototype demonstration Pilot-scale demonstration through industry connects                  | 0              | 0              | 2              | 2              | 1              | 4-5                                   |
| Digital modelling platform (kinetics + flowsheet + optimization)                                                         | 0              | 2              | 2              | 1              | 1              | 4-6                                   |
| Scale-up guidelines, TEA, LCA, and safety evaluation                                                                     | 0              | 0              | 2              | 2              | 1              | 5                                     |
| <b>Functional Materials &amp; Oil Technologies:</b>                                                                      |                |                |                |                |                |                                       |
| <b>Deliverables</b>                                                                                                      | <b>2026-27</b> | <b>2027-28</b> | <b>2028-29</b> | <b>2029-30</b> | <b>2030-31</b> | <b>Overall five-year deliverables</b> |
| New Light Harvesting molecules                                                                                           | 1              | 1              | 1              | 1-2            | 1              | 5-6                                   |
| Perovskite / Quantum dots for Energy harvesting                                                                          | 1              | 1              | 1-2            | 1              | 1-2            | 5-7                                   |
| Scale-up of Al-ion Battery                                                                                               | 1              | 1              | 1              | 1              | 1-2            | 5-6                                   |

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| Attempts to develop All-Solid-State Polymer Electrolytes                                        | 1                                                                                                                                                                                                                                                                 | 1                   | 1                   | 1                   | 1-2 | 5-6 |
| Isolation and synthesis of bioactive lipids & biolubricant base stocks.                         | 1-2                                                                                                                                                                                                                                                               | 1-2                 | 1-2                 | 1-2                 | 0   | 4-8 |
| PUFA enriched probiotics                                                                        | 1                                                                                                                                                                                                                                                                 | 1                   | 0                   | 0                   | 0   | 1-2 |
| Supercritical extraction/enrichment of lipid-soluble targeted products and pigment              | 0                                                                                                                                                                                                                                                                 | 2                   | 2                   | 0                   | 0   | 3-4 |
| Development of Novel Cationic / ionizable lipids & lipid formulations for delivery applications | 1 to 2 formulations                                                                                                                                                                                                                                               | 1 to 2 formulations | 1 to 2 formulations | 1 to 2 formulations | 0   | 4-6 |
| Identification & formulation of potent/lead bioactive lipids/saponins                           | 1-2                                                                                                                                                                                                                                                               | 1-2                 | 1-2                 | 0                   | 0   | 3-6 |
| Biolubricant formulations for industrial application                                            | 0                                                                                                                                                                                                                                                                 | 1-2                 | 1-2                 | 1-2                 | 1-2 | 4-6 |
| 2.3                                                                                             | Indicate final outcomes for the project in the form of measurable indicators which can be used for impact assessment/evaluation after the project is complete. Baseline data or survey against which such outcomes would be benchmarked should also be mentioned. |                     |                     |                     |     |     |

#### Healthcare:

| Work Package                                                                                                                                                                                                           | Proposed output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Outcome                                                                                   |
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| Phytochemical investigation medicinal plants to isolate secondary metabolites and scientific validation of traditional medicines/ medicinal plants to discover bioactive natural products or phytopharmaceutical leads | <ul style="list-style-type: none"> <li>Exploration of unexplored medicinal plants for isolation of bioactive secondary metabolites.</li> <li>Bioactivity-guided fractionation and structure elucidation of novel natural products</li> <li>Scientific validation of traditional medicines for discovery of phytopharmaceutical leads</li> <li>Marker compound identification and standardization of herbal drugs and polyherbal formulations</li> <li>Chemical fingerprinting and quality control of extracts/formulations using HPLC, HPTLC, LC-MS</li> </ul> | 1-2 phytopharmaceutical leads<br><br>3-4 new patent applications<br><br>8-10 publications |
| Design and synthesis of new molecules against therapeutic                                                                                                                                                              | <ul style="list-style-type: none"> <li>Structure-based drug design and molecular modelling for lead identification and optimization</li> <li>Synthesis of focused molecular libraries and scaffold diversification for SAR generation</li> </ul>                                                                                                                                                                                                                                                                                                               | 1-2 preclinical leads<br><br>3-4 new patent applications                                  |



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| targets and establishment of SAR, testing for ADME properties for selecting leads for in-vivo validation                                  | <ul style="list-style-type: none"> <li>Development of peptidomimetics and peptide-based therapeutics with improved stability and potency.</li> <li>Design and development of photo responsive drug delivery systems for controlled release applications</li> <li>Synthesis of fluorescent molecular probes for imaging, diagnostics, and target engagement studies</li> </ul>                                                                                                                                                                                        | 15-20 publications                                                                                                                                                                                        |
| Development of Sustainable Synthetic Approaches towards Pharmaceuticals                                                                   | <ul style="list-style-type: none"> <li>Synthesis of 60-75 potent biologically active natural products and their analogues towards the development of potent molecules including peptides, nucleosides and nucleotides.</li> <li>New libraries of therapeutically relevant new chemical entities.</li> <li>Sustainable catalytic methodologies.</li> <li>Green activation strategies using electro- &amp; photochemistry and flow-chemistry</li> <li>Scalable continuous-flow processes for Key starting materials towards Drug Substances.</li> </ul>                | 350 publications<br>25 patents<br>3000 - 4000 Chemical Entities<br>100 Ph. D Students to be trained                                                                                                       |
| Screening and Identification of potential hits for cancer therapeutics, neurotrophic activity, anti-bacterial and anti-fungal properties. | <ul style="list-style-type: none"> <li>Creating a facility for phenotypic screening of compounds from National Mol-Bank of CSIR-IICT.</li> <li>Identification of primary hits for cancer, CNS disorders and infectious diseases to enable lead discovery and structure-activity relationship.</li> </ul>                                                                                                                                                                                                                                                             | 30-40 Potential hits<br><br>100-125 publications<br>50-60 Ph. D Students to be trained                                                                                                                    |
| To assess the mode of action of lead compound and analyze the in-vivo efficacy of hit / lead compounds in animal models                   | <ul style="list-style-type: none"> <li>Analysis of the signalling mechanism / target by which the compound exerts its functional effects to understand its mode of action.</li> <li>Detailed MOA studies employing biochemical, cell biology and molecular biology tools.</li> <li>To study the bioavailability of hit compounds</li> <li>Analysis of its in vivo efficacy in appropriate disease models.</li> <li>To study the detailed PK-PD properties of lead compound.</li> <li>Generation of trained manpower in the domain of drug discovery</li> </ul>       | 5-8 Potential Leads<br><br>8-10 Patents<br><br>15-20 publications                                                                                                                                         |
| <b>Agrochemicals:</b>                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                           |
| <b>Work Package</b>                                                                                                                       | <b>Proposed output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Outcome</b>                                                                                                                                                                                            |
| Development of process technologies for market driven fluorochemicals, speciality chemicals, agrochemicals and pheromones                 | <ul style="list-style-type: none"> <li>Economically viable and industrially feasible process technologies for               <ol style="list-style-type: none"> <li>next-generation refrigerants;</li> <li>agrochemicals going off-patent</li> </ol> </li> <li>Adoption of semiochemicals in integrated pest management (IPM)</li> <li>Development of new formulation technologies</li> <li>Preparation of a designed library of compounds for agrochemical activity studies</li> <li>Integration of Artificial Intelligence (AI) in agrochemical research</li> </ul> | 100 - 150 publications<br><br>5-10 patents<br><br>Designed library of 1000 compounds for agrochemical activity evaluation<br><br>10 – 20 potential lead compounds for crop protection<br><br>20- 30 Ph. D |

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| <b>Waste to Wealth:</b>                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                   |
| <b>Work Package</b>                                                                                                                    | <b>Proposed output</b>                                                                                                                                                                                                                                                                                                                                                    | <b>Outcome</b>                                                                                    |
| Anaerobic Gas Lift Reactor (AGR) and Dry Anaerobic Digester with unique Hydrodynamics (DADUH) for the Treatment of Organic Solid Waste | <ul style="list-style-type: none"> <li>Technology for the generation of biogas and fermented organic manure from Organic waste (Anaerobic Gas Lift Reactor (AGR))</li> <li>Technology for the generation of compressed biogas (CBG) and fermented organic manure (FOM) from lignocellulosic biomass (Dry Anaerobic Digester with unique Hydrodynamics (DADUH))</li> </ul> | Technology transfers<br>2-3<br>150 publications<br>2-3 patents<br>20 Ph. D Students to be trained |
| Bi-phasic Anaerobic Digestion System for High Methane Biogas Production                                                                | <ul style="list-style-type: none"> <li>Technology for the generation of enriched methane from Organic and agriculture waste (Bi-Phasic anaerobic system)</li> </ul>                                                                                                                                                                                                       | Technology transfers<br>02<br>50 publications<br>20 Ph. D Students to be trained                  |
| Accelerated Anaerobic Composting (AAC) of Organic Waste                                                                                | <ul style="list-style-type: none"> <li>Technology for the generation of soil conditioner from green waste (Accelerated anaerobic composting)</li> </ul>                                                                                                                                                                                                                   | Technology transfers<br>02<br>50 publications<br>20 Ph. D Students to be trained                  |
| Green Hydrogen Production in Biorefinery Platform                                                                                      | <ul style="list-style-type: none"> <li>Technology for the generation of Bi-Hydrogen and volatile fatty acids from food waste</li> </ul>                                                                                                                                                                                                                                   | Technology transfers<br>2-3<br>50 publications<br>01 patents<br>20 Ph. D Students to be trained   |
| Biosuccinic acid Production through Hybrid Fermentation Route-Bioprocess Development                                                   | <ul style="list-style-type: none"> <li>Technology for Methanol production from Fruit Waste</li> </ul>                                                                                                                                                                                                                                                                     | Technology transfers<br>1-2<br>20 publications<br>10 Ph. D Students to be trained                 |
| Co-fermenting Lactic Acid and Glucose towards Caproic Acid Production                                                                  | <ul style="list-style-type: none"> <li>Technology for Biosilica, Lignin, Porous Carbon and Aerogels from Agriculture Biomass</li> </ul>                                                                                                                                                                                                                                   | Technology transfers<br>01<br>20 publications<br>01 patents<br>05 Ph. D Students to be trained    |
| <b>Specialty Chemicals:</b>                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                   |
| Development of Advanced Process Technologies for Specialty Chemicals Synthesis                                                         | <ul style="list-style-type: none"> <li>Development and pilot-scale demonstration of optimized batch, semi-batch, and continuous process technologies for at least five priority specialty chemicals, including validated reaction pathways, catalyst compatibility, and quantified performance data (yield, selectivity, energy efficiency, and stability).</li> </ul>    | Publications: 30<br>Patents: 1-2<br>PhDs: 4-6<br>Technology transfers: 3-5                        |
| Process Intensification for established Chemistry and                                                                                  | <ul style="list-style-type: none"> <li>Development and demonstration of modular intensified reactor prototypes and advanced continuous reactor platforms (e.g., customized flow, oscillatory baffled, reactive distillation, hybrid, and membrane-assisted) with validated</li> </ul>                                                                                     | Publications: 30<br>Patents: 1-2<br>PhDs: 4-6                                                     |

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| Modular Reactor Engineering                                                                                            | improvements in mixing, heat/mass transfer, and integrated reaction–separation performance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Technology transfers: 1-2                                                 |
| Know-How Development, Modelling Platforms, and Scale-Up Methodologies                                                  | <ul style="list-style-type: none"> <li>An integrated process know-how package including catalyst design and screening data, a validated digital modelling platform for kinetic and flowsheet simulation, and comprehensive scale-up guidelines supported by TEA, LCA, and safety assessments.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                  | Publications: 20<br>Patents: 1<br>PhDs: 4-6<br>Technology transfers: 1-2  |
| <b>Carbon Capture, Utilization &amp; Storage (CCUS):</b>                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                           |
| <b>Description</b>                                                                                                     | <b>Outputs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Outcomes</b>                                                           |
| Carbon Capture (CC)                                                                                                    | <ul style="list-style-type: none"> <li>Development and screening of advanced phase-change amine solvent systems for efficient CO<sub>2</sub> absorption.</li> <li>Design and validation of intensified gas–liquid contactors (membrane contactors, rotating packed beds).</li> <li>Lab- and pilot-scale CO<sub>2</sub> capture modules demonstrating improved mass transfer and compact system design.</li> <li>Optimized solvent blends with improved absorption capacity, stability, and cyclic performance.</li> <li>Integrated process modelling and simulation (CFD/Aspen) for absorber–stripper systems.</li> <li>Techno-economic analysis (TEA) and life cycle assessment (LCA) for process evaluation.</li> </ul> | Publications: 10<br>Patents: 2<br>PhDs:3<br>Technology transfers: 1-2     |
| Carbon Utilization (CU)                                                                                                | <ul style="list-style-type: none"> <li>Development of stable catalytic materials for CO<sub>2</sub> conversion reactions.</li> <li>Design and testing of intensified catalytic reactors for CO<sub>2</sub> transformation.</li> <li>Demonstration of lab- and pilot-scale catalytic conversion systems.</li> <li>Optimization of catalysts for production of hydrogen-rich syngas, dimethyl ether (DME) and other value-added products</li> <li>Kinetic studies, reactor modelling, and integrated process simulations.</li> <li>TEA and LCA evaluation for CO<sub>2</sub> utilization pathways.</li> </ul>                                                                                                               | Publications: 8-10<br>Patents: 1-2<br>PhDs:3<br>Technology transfers: 1-2 |
| <b>Functional Materials &amp; Oil Technologies:</b>                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                           |
| <b>Work Package</b>                                                                                                    | <b>Proposed Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Outcome</b>                                                            |
| Development of advanced functional materials and composites for efficient energy harvesting, and storage applications. | <ul style="list-style-type: none"> <li>Metal-free organic NIR light-harvesters</li> <li>Hole transporting materials (HTMs) for perovskite solar cells (PSCs)</li> <li>Electrodes and electrolytes for Li ion Rechargeable Battery</li> <li>Vegetable oil-based nutraceutical</li> <li>Novel bioactive lipids and pigments</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      | 50-60 Publications<br>2-4 patents<br>15-18 PhD students                   |
| Value addition to the by-product of the edible oil refining industry and silk industry.                                | <ul style="list-style-type: none"> <li>Vegetable oil-based nutraceuticals</li> <li>PUFA enriched probiotics</li> <li>Novel bioactive lipids and saponins</li> <li>Innovative delivery system</li> <li>Enriched protein from desilked pupae</li> <li>Biolubricant base stocks / formulations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                    | 20-30 Publications<br>1-2 patents<br>8-10 PhD students                    |
| <b>3.</b>                                                                                                              | <b>Project Cost</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                           |
| 3.1                                                                                                                    | Cost Estimates for the project along with scheduled duration (both year and activity wise). Also, the basis for these cost estimates along with reference dates for normative costing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Reference Date<br>January 2026<br>Please See Sl. No. 22(C)                |

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| 3.2       | In case land is to be acquired, the details of land cost, including cost of rehabilitation/ resettlement needs to be provided                                                                                     | Not Applicable                                                                                                                                                                                                                                                                |
| 3.3       | In case pre-investment activities are required, how much is proposed to be spent on these, with details activity-wise?                                                                                            | Not Applicable                                                                                                                                                                                                                                                                |
| 3.4       | Whether price escalation during the project time cycle has been included in the cost estimates and at what rates?                                                                                                 | Not Included                                                                                                                                                                                                                                                                  |
| 3.5       | Whether the Project involves any foreign exchange element, the provision made or likely impact of exchange rate risks?                                                                                            | Involves. Provisions made within the budget                                                                                                                                                                                                                                   |
| 3.6       | In case of the Revised Cost Estimates, a variation analysis along with the Report of the Revised Cost Committee needs to be attached.                                                                             | Not Applicable                                                                                                                                                                                                                                                                |
| <b>4</b>  | <b>Project Finance</b>                                                                                                                                                                                            |                                                                                                                                                                                                                                                                               |
| 4.1       | Indicate the sources of project finance: budgetary support, internal and extra-budgetary sources, external aid, etc.                                                                                              | Under National Laboratories Scheme-Central Sector Scheme, Sub-Scheme: Unit and Laboratory Inhouse Projects (ULIP)                                                                                                                                                             |
| 4.2       | Indicate the cost components, if any, that will be shared by the state governments, local bodies, user beneficiaries or private parties?                                                                          | 100% Centrally Funded                                                                                                                                                                                                                                                         |
| 4.3       | In case of funding from internal and extra-budgetary resources, availability of internal resources may be supported by projections and their deployment on other projects?                                        | Not Applicable                                                                                                                                                                                                                                                                |
| 4.4       | Please indicate funding tie-ups for the loan components, if any, both domestic and foreign, along-with terms and conditions of loan based on consent/comfort letters.                                             | Not Applicable                                                                                                                                                                                                                                                                |
| 4.5       | If government support/loan is intended, it may be indicated whether such funds have been tied up?                                                                                                                 | Proposed under the National Laboratories Scheme under the 16 FCC                                                                                                                                                                                                              |
| 4.6       | Please provide the leveraging details, including debt-equity and Interest coverage ratios, along with justification for the same.                                                                                 | Not Applicable                                                                                                                                                                                                                                                                |
| 4.7       | Mention the legacy arrangements after the project is complete, in particular, arrangements for the maintenance and upkeep of assets that will be created?                                                         | Responsibility of the constituent laboratory of the CSIR, where the project is implemented.                                                                                                                                                                                   |
| <b>5.</b> | <b>Project Viability</b>                                                                                                                                                                                          |                                                                                                                                                                                                                                                                               |
| 5.1       | For projects which have identifiable stream of financial returns, the financial internal rate of return may be calculated. The hurdle rate will be considered at 10 percent.                                      | Direct Financial return is in terms of technology transfer, consultancy services, testing services and Indirect Financial returns are on the basis of Intellectual knowledge, publications, and IP.                                                                           |
| 5.2       | In case of projects with identifiable economic returns, the economic rate of return may be calculated. In such cases project viability will be determined by taking both financial and economic returns together. | The project addresses to the fundamental aspects of self-reliance and also S&T for improving quality of life in the country. There is a likely economic return from the project, possibly after the project period. At the present moment this has not been quantified fully. |

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| 5.3                                                                                                                                                                                                                    | In case of proposals where both financial and economic returns are not readily quantifiable, the measurable benefits/outcomes simply may be indicated.                                                                                                                                                                                                                                                                                                                                                                                                                                          | The measurable outputs and outcomes from the project are detailed in Sl. No. 23 of the DPR attached as well as the same is given below |
| <b>Healthcare:</b>                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                        |
| <b>Work Package</b>                                                                                                                                                                                                    | <b>Proposed output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Outcome</b>                                                                                                                         |
| Phytochemical investigation medicinal plants to isolate secondary metabolites and scientific validation of traditional medicines/ medicinal plants to discover bioactive natural products or phytopharmaceutical leads | <ul style="list-style-type: none"> <li>• Exploration of unexplored medicinal plants for isolation of bioactive secondary metabolites.</li> <li>• Bioactivity-guided fractionation and structure elucidation of novel natural products</li> <li>• Scientific validation of traditional medicines for discovery of phytopharmaceutical leads</li> <li>• Marker compound identification and standardization of herbal drugs and polyherbal formulations chemical fingerprinting and quality control of extracts/formulations using HPLC, HPTLC, LC-MS</li> </ul>                                   | <p>1-2 phytopharmaceutical leads</p> <p>3-4 new patent applications</p> <p>8-10 publications</p>                                       |
| Design and synthesis of new molecules against therapeutic targets and establishment of SAR, testing for ADME properties for selecting leads for in-vivo validation                                                     | <ul style="list-style-type: none"> <li>• Structure-based drug design and molecular modelling for lead identification and optimization</li> <li>• Synthesis of focused molecular libraries and scaffold diversification for SAR generation</li> <li>• Development of peptidomimetics and peptide-based therapeutics with improved stability and potency.</li> <li>• Design and development of photo responsive drug delivery systems for controlled release applications</li> <li>• Synthesis of fluorescent molecular probes for imaging, diagnostics, and target engagement studies</li> </ul> | <p>1-2 preclinical leads</p> <p>3-4 new patent applications</p> <p>15-20 publications</p>                                              |
| Development of Sustainable Synthetic                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>• Synthesis of 60-75 potent biologically active natural products and their analogues towards the</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>350 publications</p> <p>25 patents</p>                                                                                              |
| Approaches towards Pharmaceuticals                                                                                                                                                                                     | <p>development of potent molecules including peptides, nucleosides and nucleotides.</p> <ul style="list-style-type: none"> <li>• New libraries of therapeutically relevant new chemical entities.</li> <li>• Sustainable catalytic methodologies.</li> <li>• Green activation strategies using electro- &amp; photochemistry and flow-chemistry</li> <li>• Scalable continuous-flow processes for Key starting materials towards Drug Substances.</li> </ul>                                                                                                                                    | <p>3000 - 4000 Chemical Entities</p> <p>100 Ph. D Students to be trained</p>                                                           |
| Screening and Identification of potential hits for cancer therapeutics, neurotrophic activity, anti-bacterial and anti-fungal properties.                                                                              | <ul style="list-style-type: none"> <li>• Creating a facility for phenotypic screening of compounds from National Mol-Bank of CSIR-IICT.</li> <li>• Identification of primary hits for cancer, CNS disorders and infectious diseases to enable lead discovery and structure-activity relationship.</li> </ul>                                                                                                                                                                                                                                                                                    | <p>30-40 Potential hits</p> <p>100-125 publications</p> <p>50-60 Ph. D Students to be trained</p>                                      |

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| To assess the mode of action of lead compound and analyze the in-vivo efficacy of hit / lead compounds in animal models                | <ul style="list-style-type: none"> <li>• Analysis of the signalling mechanism / target by which the compound exerts its functional effects to understand its mode of action.</li> <li>• Detailed MOA studies employing biochemical, cell biology and molecular biology tools.</li> <li>• To study the bioavailability of hit compounds</li> <li>• Analysis of its in vivo efficacy in appropriate disease models.</li> <li>• To study the detailed PK-PD properties of lead compound.</li> <li>• Generation of trained manpower in the domain of drug discovery</li> </ul>           | 5-8 Potential Leads<br><br>8-10 Patents<br><br>15-20 publications                                                                                                                                         |
| <b>Agrochemicals:</b>                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                           |
| <b>Work Package</b>                                                                                                                    | <b>Proposed output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Outcome</b>                                                                                                                                                                                            |
| Development of process technologies for market driven fluoroc hemicals, speciality chemicals, agrochemicals and pheromones             | <ul style="list-style-type: none"> <li>• Economically viable and industrially feasible process technologies for               <ul style="list-style-type: none"> <li>a) next-generation refrigerants;</li> <li>b) agrochemicals going off-patent</li> </ul> </li> <li>• Adoption of semiochemicals in integrated pest management (IPM)</li> <li>• Development of new formulation technologies</li> <li>• Preparation of a designed library of compounds for agrochemical activity studies</li> <li>• Integration of Artificial Intelligence (AI) in agrochemical research</li> </ul> | 100 - 150 publications<br><br>5-10 patents<br><br>Designed library of 1000 compounds for agrochemical activity evaluation<br><br>10 – 20 potential lead compounds for crop protection<br><br>20- 30 Ph. D |
| <b>Waste to Wealth:</b>                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                           |
| <b>Work Package</b>                                                                                                                    | <b>Proposed output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Outcome</b>                                                                                                                                                                                            |
| Anaerobic Gas Lift Reactor (AGR) and Dry Anaerobic Digester with unique Hydrodynamics (DADUH) for the Treatment of Organic Solid Waste | <ul style="list-style-type: none"> <li>• Technology for the generation of biogas and fermented organic manure from Organic waste (Anaerobic Gas Lift Reactor (AGR))</li> <li>• Technology for the generation of compressed biogas (CBG) and fermented organic manure (FOM) from lignocellulosic biomass (Dry Anaerobic Digester with unique Hydrodynamics (DADUH))</li> </ul>                                                                                                                                                                                                        | Technology transfers 2-3<br>150 publications<br>2-3 patents<br>20 Ph. D Students to be trained                                                                                                            |
| Bi-phasic Anaerobic Digestion System for High Methane Biogas Production                                                                | <ul style="list-style-type: none"> <li>• Technology for the generation of enriched methane from Organic and agriculture waste (Bi-Phasic anaerobic system)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                | Technology transfers 02<br>50 publications<br>20 Ph. D Students to be trained                                                                                                                             |
| Accelerated Anaerobic Composting (AAC) of Organic Waste                                                                                | <ul style="list-style-type: none"> <li>• Technology for the generation of soil conditioner from green waste (Accelerated anaerobic composting)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                            | Technology transfers 02<br>50 publications<br>20 Ph. D Students to be trained                                                                                                                             |
| Green Hydrogen Production in Biorefinery Platform                                                                                      | <ul style="list-style-type: none"> <li>• Technology for the generation of Bi-Hydrogen and volatile fatty acids from food waste</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            | Technology transfers 2-3<br>50 publications<br>01 patents<br>20 Ph. D Students to be trained                                                                                                              |
| Biosuccinic acid Production through Hybrid Fermentation Route- Bioprocess Development                                                  | <ul style="list-style-type: none"> <li>• Technology for Methanol production from Fruit Waste</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Technology transfers 1-2<br>20 publications<br>10 Ph. D Students to be trained                                                                                                                            |

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| Co-fermenting Lactic Acid and Glucose towards Caproic Acid Production             | <ul style="list-style-type: none"> <li>Technology for Biosilica, Lignin, Porous Carbon and Aerogels from Agriculture Biomass</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Technology transfers 01<br>20 publications<br>01 patents<br>05 Ph. D Students to be trained |
| <b>Specialty Chemicals:</b>                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                             |
| <b>Work package</b>                                                               | <b>Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Outcome</b>                                                                              |
| Development of Advanced Process Technologies for Specialty Chemicals Synthesis    | <ul style="list-style-type: none"> <li>Development and pilot-scale demonstration of optimized batch, semi-batch, and continuous process technologies for at least five priority specialty chemicals, including validated reaction pathways, catalyst compatibility, and quantified performance data (yield, selectivity, energy efficiency, and stability).</li> </ul>                                                                                                                                                                                                                                                    | Publications: 30<br>Patents: 1-2<br>PhDs: 4-6<br>Technology transfers: 3-5                  |
| Process Intensification for established Chemistry and Modular Reactor Engineering | <ul style="list-style-type: none"> <li>Development and demonstration of modular intensified reactor prototypes and advanced continuous reactor platforms (e.g., customized flow, oscillatory baffled, reactive distillation, hybrid, and membrane-assisted) with validated improvements in mixing, heat/mass transfer, and integrated reaction–separation performance.</li> </ul>                                                                                                                                                                                                                                         | Publications: 30<br>Patents: 1-2<br>PhDs: 4-6<br>Technology transfers: 1-2                  |
| Know-How Development, Modelling Platforms, and Scale-Up Methodologies             | <ul style="list-style-type: none"> <li>An integrated process know-how package including catalyst design and screening data, a validated digital modelling platform for kinetic and flowsheet simulation, and comprehensive scale-up guidelines supported by TEA, LCA, and safety assessments.</li> </ul>                                                                                                                                                                                                                                                                                                                  | Publications: 20<br>Patents: 1<br>PhDs: 4-6<br>Technology transfers: 1-2                    |
| <b>Carbon Capture, Utilization &amp; Storage (CCUS):</b>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                             |
| <b>Description</b>                                                                | <b>Outputs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Outcomes</b>                                                                             |
| Carbon Capture (CC)                                                               | <ul style="list-style-type: none"> <li>Development and screening of advanced phase-change amine solvent systems for efficient CO<sub>2</sub> absorption.</li> <li>Design and validation of intensified gas–liquid contactors (membrane contactors, rotating packed beds).</li> <li>Lab- and pilot-scale CO<sub>2</sub> capture modules demonstrating improved mass transfer and compact system design.</li> <li>Optimized solvent blends with improved absorption capacity, stability, and cyclic performance.</li> <li>Integrated process modelling and simulation (CFD/Aspen) for absorber–stripper systems.</li> </ul> | Publications: 10<br>Patents: 2<br>PhDs:3<br>Technology transfers: 1-2                       |
|                                                                                   | <ul style="list-style-type: none"> <li>Techno-economic analysis (TEA) and life cycle assessment (LCA) for process evaluation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                             |
| Carbon Utilization (CU)                                                           | <ul style="list-style-type: none"> <li>Development of stable catalytic materials for CO<sub>2</sub> conversion reactions.</li> <li>Design and testing of intensified catalytic reactors for CO<sub>2</sub> transformation.</li> <li>Demonstration of lab- and pilot-scale catalytic conversion systems.</li> <li>Optimization of catalysts for production of hydrogen-rich syngas, dimethyl ether (DME) and other value-added products</li> <li>Kinetic studies, reactor modelling, and integrated process simulations.</li> <li>TEA and LCA evaluation for CO<sub>2</sub> utilization pathways.</li> </ul>               | Publications: 8-10<br>Patents: 1-2<br>PhDs:3<br>Technology transfers: 1-2                   |
| <b>Functional Materials &amp; Oil Technologies:</b>                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                             |

| Work Package                                                                                                           | Proposed Output                                                                                                                                                                                                                                                                                                                                                                            | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| Development of advanced functional materials and composites for efficient energy harvesting, and storage applications. | <ul style="list-style-type: none"> <li>• Metal-free organic NIR light-harvesters</li> <li>• Hole transporting materials (HTMs) for perovskite solar cells (PSCs)</li> <li>• Electrodes and electrolytes for Li ion Rechargeable Battery</li> <li>• Vegetable oil-based nutraceutical</li> <li>• Novel bioactive lipids and pigments</li> </ul>                                             | 50-60 Publications<br>2-4 patents<br>15-18 PhD students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Value addition to the by-product of the edible oil refining industry and silk industry.                                | <ul style="list-style-type: none"> <li>• Vegetable oil-based nutraceuticals</li> <li>• PUFA enriched probiotics</li> <li>• Novel bioactive lipids and saponins</li> <li>• Innovative delivery system</li> <li>• Enriched protein from desilked pupae</li> <li>• Biolubricant base stocks / formulations</li> </ul>                                                                         | 20-30 Publications<br>1-2 patents<br>8-10 PhD students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>6</b>                                                                                                               | <b>Approvals and Clearances</b>                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                                                                        | Requirement of mandatory approvals/clearances of various local, state and national bodies and their availability may be indicated in a tabular form (land acquisition, environment, forestry, wildlife etc.) In case land is required, it may be clearly mentioned whether the land is in the possession of the agency, free from encumbrances, or encroached or stuck in legal processes. | <p><b>Healthcare:</b> Healthcare &amp; Drug Discovery: Compliance with national drug discovery and preclinical research regulations, including CPCSEA guidelines for animal studies. IP protection ensured through timely national/international patent filings, governed by MTAs, CDAs, and collaborative agreements.</p> <p><b>Agrochemicals:</b> Professional IP analysis conducted with URDIP to identify innovation opportunities. Active molecules and process developments systematically evaluated for patentability, with</p> |



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|  |  | <p>appropriate filings to safeguard commercialization potential.</p> <p><b>Waste to Wealth:</b> Governed by Solid Waste Management Rules (2026), MNRE Waste-to-Energy Programme, and CPCB/SPCB guidelines. Projects require MNRE approval, strict emission compliance, and adherence to PPP competitive bidding norms. Extended Producer Responsibility (EPR) and progressive recycling targets are mandatory.</p> <p><b>Specialty Chemicals:</b> Processes designed to meet environmental regulations and avoid compliance issues. Results published, with patents filed for commercialization. Partnerships governed by CSIR guidelines through MTAs, CDAs, and collaborative agreements.</p> <p><b>Carbon Capture, Utilization &amp; Storage (CCUS):</b> Work aligned with environmental, safety, and industrial regulations for chemical processing and emissions. Techno-economic and environmental evaluations ensure compliance, safe operation, and long-term viability.</p> <p><b>Functional Materials &amp; Oil Technologies:</b> Implemented under CSIR-IICT jurisdiction, following CSIR guidelines for third-party involvement, technology transfer, and commercialization. IP planning, documentation, and regulatory compliance integrated throughout.</p> |
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| Optimization of parameters of supercritical extraction for efficient and selective extraction of targeted products. |                                                                                                                                                                                                                                   |                                                                                                      |  |  |  |  |  |  |  |
| Development of Novel Cationic / ionizable lipids & lipid formulations for delivery applications                     |                                                                                                                                                                                                                                   |                                                                                                      |  |  |  |  |  |  |  |
| Identification & formulation of potent/lead bioactive lipids/saponins                                               |                                                                                                                                                                                                                                   |                                                                                                      |  |  |  |  |  |  |  |
| Scale-up of lead products, techno-economic assessment and devising roadmap for commercialization.                   |                                                                                                                                                                                                                                   |                                                                                                      |  |  |  |  |  |  |  |
| 8.4                                                                                                                 | Please indicate the monitoring framework, including MIS, and the arrangements for internal/statutory audit.                                                                                                                       | Half Yearly Monitoring by Research Council of the Laboratory                                         |  |  |  |  |  |  |  |
| 8.5                                                                                                                 | Please indicate what arrangements have been made for impact assessment after the project is complete?                                                                                                                             | Third Party Impact Assessment is Planned after the 4 <sup>th</sup> year of the project               |  |  |  |  |  |  |  |
| 9                                                                                                                   | <b>Comments of the Financial Advisor</b> , NITI Aayog, Department of Expenditure and other Ministries/Departments may be summarized in tabular form along with how they have been internalized and used to improve this proposal. |                                                                                                      |  |  |  |  |  |  |  |
| 10                                                                                                                  | <b>Approval Sought:</b>                                                                                                                                                                                                           |                                                                                                      |  |  |  |  |  |  |  |
|                                                                                                                     |                                                                                                                                                                                                                                   | (-----)<br>Joint Secretary to the Government of India<br>Tel. No .. ____<br>Fax No .. ____<br>Email. |  |  |  |  |  |  |  |
|                                                                                                                     | DPR is attached.                                                                                                                                                                                                                  |                                                                                                      |  |  |  |  |  |  |  |

CHECKLIST FOR FINANCIAL APPRAISAL AND APPROVAL OF THE PROJECT: **IICT-CS**

|              |                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                        |             |
|--------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1            | Project Title                                                                                          | Chemical Sciences and Technologies for Health, Agriculture and Environment (IICT-CS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                        |             |
| 2            | Project Code                                                                                           | IICT- CS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                        |             |
| 3            | Laboratory                                                                                             | CSIR-IICT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                        |             |
| 4            | Total Outlay Sought and Tentatively Suggested as per EFC                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                        |             |
| 5            | Recurring Expenditure Sought and Tentatively Suggested as per EFC                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                        |             |
| 6            | Capital Expenditure Sought and Tentatively Suggested as per EFC                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                        |             |
| 7            | Overview and Justification                                                                             | <p>The proposed budget over five years is strategically aligned with the project's multidisciplinary objectives in healthcare, agrochemicals, waste-to-wealth, CCUS, and advanced materials. The allocation balances capital investment in state-of-the-art equipment and recurring costs for consumables, manpower, and operations, ensuring end-to-end research capability from discovery to pilot-scale validation. Significant provisions for skilled human resources and advanced infrastructure are essential to deliver high-impact outputs including patents, publications, and industry-ready technologies. The budget is justified by its potential to drive national priorities such as self-reliance, sustainability, and industrial competitiveness.</p> |                                                                                                                                                                                                                                                                                                                                                                        |             |
| 8            | Justification for Capital Expenditure                                                                  | Provided against each equipment in below table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                        |             |
| <b>S. No</b> | <b>Name of the Equipment/ No.</b>                                                                      | <b>Estimated cost in lakhs (INR)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Justification (in 2 points)</b>                                                                                                                                                                                                                                                                                                                                     | <b>Year</b> |
| 1.           | Biochemistry Analyzer (Integrated clinical chemistry and immunoassay analyzer) 1 No. (Applied Biology) | 60 Lakhs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | This instrument is essential for day-to-day research work involving biochemical analysis of serum samples from laboratory animals as a part of research in the areas of inflammation, cancer and metabolic disorders including diabetes, NASH, etc.                                                                                                                    | 2026-27     |
| 2.           | Real-Time PCR system (96 and 384 well module) with computer 1 No. (Applied Biology)                    | 20 Lakhs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Acquiring an RT-qPCR machine is crucial for precise, high-throughput gene expression analysis in <i>in vitro</i> and <i>in vivo models</i> . It supports studies on disease mechanisms, therapeutic responses, and biomarker discovery with high sensitivity and specificity. This versatile tool enhances the lab's capacity for advanced molecular biology research. | 2026-27     |
| 3.           | Microplate reader 1 No. (Applied Biology)                                                              | 20 Lakhs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | A microplate reader is vital for high-throughput analysis of biochemical and cellular assays, offering versatile measurements of absorbance, fluorescence, and luminescence.                                                                                                                                                                                           | 2026-27     |
| 4.           | ARC Upgradation 1 No. (CEPT)                                                                           | 60 Lakhs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Accelerating Rate Calorimetry (ARC) specializes in simulating worst-case adiabatic (no heat loss) conditions to study thermal runaway and hazards of self-accelerating exothermic reactions, unlike standard reaction calorimeters (like RC 1 or DSC),                                                                                                                 | 2026-27     |

|     |                                                                                                                                                            |           |                                                                                                                                                                                                                                                                                                                                                                       |          |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|     |                                                                                                                                                            |           | which might operate under controlled cooling/isothermal conditions to measure heat flow for process optimization.                                                                                                                                                                                                                                                     |          |
| 5.  | Melt Extrusion Unit<br>1 No. (CEPT)                                                                                                                        | 50 Lakhs  | For the GFL project to extrude ion exchange membranes by melt extrusion at IICT                                                                                                                                                                                                                                                                                       | 2026-27  |
| 6.  | 1-5 Kg Catalyst preparation units (Driers, Extruders, glass reactors, Low temperature cooling circulators, Furnaces)<br>1 No. (Catalysis & Fine Chemicals) | 100 Lakhs | Essential for the scale up of catalysts related industrial application Required for pilot scale studies of the industry sponsored projects                                                                                                                                                                                                                            | 2026-27  |
| 7.  | Scanning Electron Microscope<br>1 No. (A&SC)                                                                                                               | 175 Lakhs | It is one of the essential equipment for material research.                                                                                                                                                                                                                                                                                                           | 2026-27  |
| 8.  | BCU-02 with 4 mm CPMAS/HRMAS probe and autosampler 1 No. (A&SC)                                                                                            | 250 Lakhs | To cater the needs of internal as well as external samples, a solid-state NMR probe is required to integrate with existing 500 MHz NMR spectrometer. This probe is also suitable for analysing Hydrogels, Resins, Insoluble Polymers and Pharmaceutical Polymers. For better utilization of the instruments, an autosampler is required.                              | 2026-27  |
| 9.  | Differential Scanning Calorimeter (DSC) and Modulated DSC 1 No. (A&SC)                                                                                     | 90 Lakhs  | We are in requirement of DSC along with MDSC.                                                                                                                                                                                                                                                                                                                         | 2026-27  |
| 10. | Ash Fusion Tester<br>1 No. (DEEE)                                                                                                                          | 75 Lakhs  | An Ash Fusion Tester is a specialized laboratory device used in coal analysis to determine the melting behaviour of coal ash at high temperatures. It is used to assess the temperature at which ash from coal starts to soften, fuse, or form a liquid, which is critical for understanding its behaviour during combustion in power plants and industrial furnaces. | 2026-27  |
| 11. | Dispersion Stability Analyzer<br>1 No. (FAC)                                                                                                               | 52 Lakhs  | Formulation Equipment                                                                                                                                                                                                                                                                                                                                                 | 2026-27  |
| 12. | Particle Size Analyzer & Zeta Potential 1 No. (FAC)                                                                                                        | 45 Lakhs  | Formulation Equipment                                                                                                                                                                                                                                                                                                                                                 | 2026-27  |
| 13. | Schrodinger Software ( <i>license for 5 years – 2026-2030</i> )<br>1 No. (NPMC)                                                                            | 25 Lakhs  | DrugDesign/Molecular modeling activities;<br>Software license is categorized under capital head.                                                                                                                                                                                                                                                                      | 2026 -27 |
| 14. | Circular Dichroism (CD)<br>1 No. (NPMC)                                                                                                                    | 70 Lakhs  | For absolute stereochemistry of isolated natural products; to get valuable information of peptides, proteins and other chromophoric chiral molecules structural conformation. Protein-ligand interaction, Drug-DNA interaction and their subsequent change in the conformation, protein folding and aggregation studies can be studied in CD spectroscopy.            | 2026-27  |
| 15. | Pulsed NMR for Solid Fat Compositions (DOLST)                                                                                                              | 50 Lakhs  | This equipment is used to measure the solid fat content of fat samples at                                                                                                                                                                                                                                                                                             | 2026-27  |



|    |                                                          |           |                                                                                                                                                                                                                                                                                                           |          |
|----|----------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|    | 1 No.                                                    |           | different temperatures. The equipment we had in our department, earned considerable amount of revenues and we are getting many enquiries from the industries. The machine is not working now and its condition is beyond repairing.                                                                       |          |
| 16 | Deep Freezer<br>-80 °C freezer 1 No. (DOLST)             | 15 Lakhs  | Dedicated deep freezer for keeping tissue culture materials to be used in tissue culture lab (under development) in OLST to preserve tissue samples                                                                                                                                                       | 2026-27  |
| 17 | Kjeldahl Equipment for Nitrogen Estimation 1 No. (DOLST) | 20 Lakhs  | This is an equipment for Protein related work. As, we have some protein-based research projects and planned basic research on protein based                                                                                                                                                               | 2026-27  |
| 18 | Gas generator (DOLST)                                    | 30 Lakhs  | Generation of H <sub>2</sub> , N <sub>2</sub> and air for GC. Being an NABL accredited lab, gas generator is indispensable for uninterrupted operation of GCs.                                                                                                                                            | 2026-27  |
| 19 | FT-IR 2<br>No. (1-<br>OSPC; 01<br>PFM)                   | 70 Lakhs  | To characterize a compound for replacement of the existing one are older (more than 15 years)                                                                                                                                                                                                             | 2026-27  |
| 20 | GC-MS 1<br>No. (OSPC)                                    | 75 Lakhs  | To characterize and monitor a chemical reaction and mainly for mission mode projects. For replacement of the existing one are older (more than 15 years)                                                                                                                                                  | 2026-27  |
| 21 | Salt-spray unit 1<br>No. (PFM)                           | 20 Lakhs  | Salt-spray unit is important and routinely required for testing accelerated corrosion performance of coating and paint formulations received from both PFM and Industry. This unit at PFM is now more than 20 yrs old and spare parts are not available and therefore not in fully functional conditions. | 2026 -27 |
| 22 | QUV weathering 1<br>No. (PFM)                            | 30 Lakhs  | Used for measuring the accelerated weathering performance of the coating and paints it will be employed for testing samples from the IICT and industry. This unit at PFM is now more than 20 yrs old and spare parts are not available and therefore not in fully functional conditions                   | 2026 -27 |
| 23 | Minor Equipment's                                        | 150 Lakhs | Stirrers, rotary evaporators, vacuum pumps, chillers, safety cabinets, Ice flake machines                                                                                                                                                                                                                 | 2026-27  |

|    |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Justification for Recurring Expenditure | <p>The recurring expenditure is essential for the effective execution and continuity of the proposed research activities under the project. A major component is allocated towards chemicals, consumables, and laboratory reagents, which are critical for conducting experiments across healthcare, agrochemicals, CCUS, and materials research domains. Additionally, technical manpower costs are necessary to support intensive R&amp;D, data generation, and technology development activities.</p> <p>Provision for travel (TA) ensures collaboration, field validation, and industry engagement, while Other Recurring Expenditure (ORE) supports maintenance, utilities, and operational requirements of sophisticated research infrastructure. Given the multidisciplinary nature and scale of deliverables including publications, patents, pilot-scale technologies, and trained manpower the recurring expenditure is fully justified to sustain high-quality research output and ensure timely achievement of project objectives.</p>                                  |
| 10 | Deliverables                            | <p>The key deliverables of the IICT-CS project include the discovery and development of novel therapeutic molecules and sustainable drug processes to strengthen pharmaceutical pipeline. It will also deliver cost-effective, environmentally safe agrochemical and fluorochemical technologies, alongside indigenous specialty chemical processes to reduce import dependence. Integrated waste-to-wealth solutions will be developed to convert solid and liquid wastes into renewable energy (biogas, bio-hydrogen, ethanol, methanol) and value-added materials like silica, lignin, and aerogels. The project will also provide scalable CO<sub>2</sub> capture and utilization systems, converting emissions into fuels and chemicals. Additionally, advanced functional materials for solar energy, batteries, and oil technologies will be created, supporting clean energy, circular economy, and national self-reliance goals. Collectively, these deliverables aim to strengthen industrial competitiveness, environmental sustainability, and societal well-being.</p> |
| 11 | Checks for Approval                     | <p>Research Council has approved the proposal and appended below.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |